

WASP-10b

R-1182

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_210105 · created 2026-07-31T04:38:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459219.6193 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.012
Transit depth (Rp/Rs)^2	3.92 +/- 0.48 [%]
Orbital Inclination (inc)	88.51 +/- 0.77
Airmass coefficient 1 (a1)	1.024 +/- 0.016
Airmass coefficient 2 (a2)	-0.0097 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#3 - [383, 187]
Optimal Aperture	3.67
Optimal Annulus	10.06
Transit Duration (day)	0.0871 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.012 vs 0.1592 (deviation 2.86 σ)
Duration fit vs published	0.0871 d vs 0.0946 d (deviation 3.01 σ)
Mid-time O-C	-10.2 min
Depth SNR	14.6
Residual scatter	1.6 %

Light curve

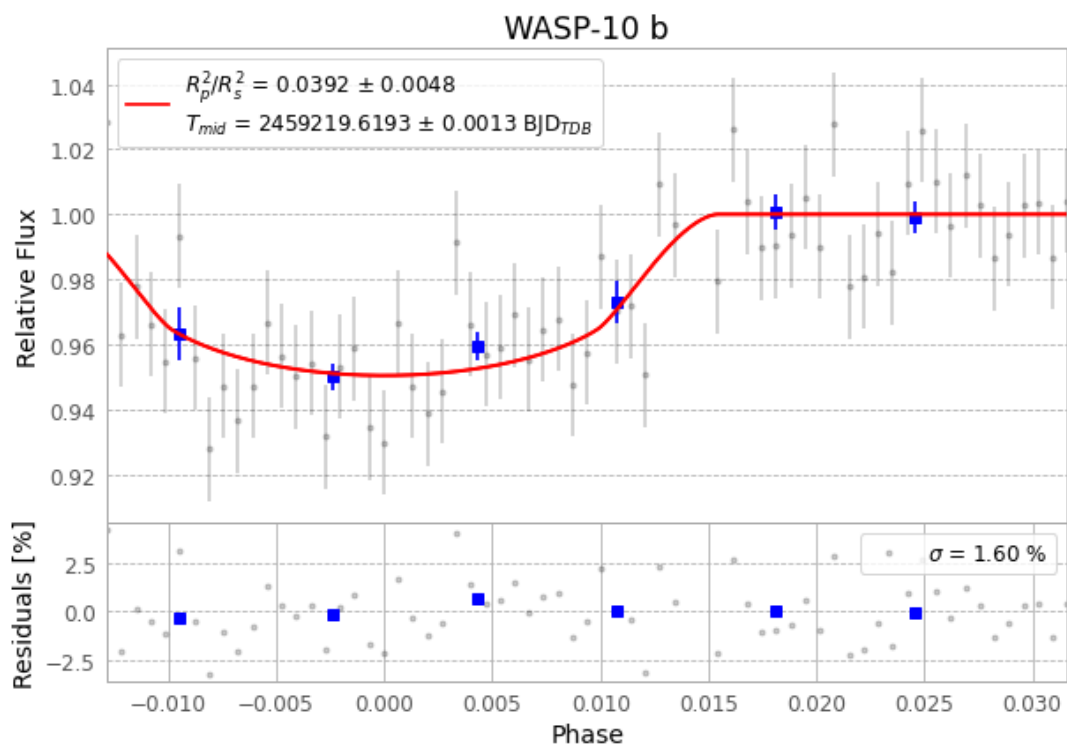


Figure 1 — FinalLightCurve_WASP-10 b_2021-01-04.png (SHA-256 57542c92cb6c217f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6098e917b5f45d1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), WASP-10210105015412.FITS → WASP-10210105051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-210105.log (34c3fee4caf5...), FinalLightCurve_WASP-10 b_2021-01-04.png (57542c92cb6c...), FinalLightCurve_WASP-10 b_2021-01-04.csv (55e4589c8096...)

Compute resources

Wall time 220.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:38Z.